List of Meetings:

- April 1st, 2025 - Zoom - Emily, Azucena, Timothy, Jacob
- April 2nd, 2025 - Zoom - Emily, Azucena, Timothy, Jacob
- April 24th, 2025 - Davis Library - Emily, Azucena, Timothy, Jacob
- April 25th, 2025 - Greenlaw - Emily, Azucena, Timothy, Jacob
- May 4th, 2025 - Undergrad Library - Emily, Jacob, Azucena, Timothy

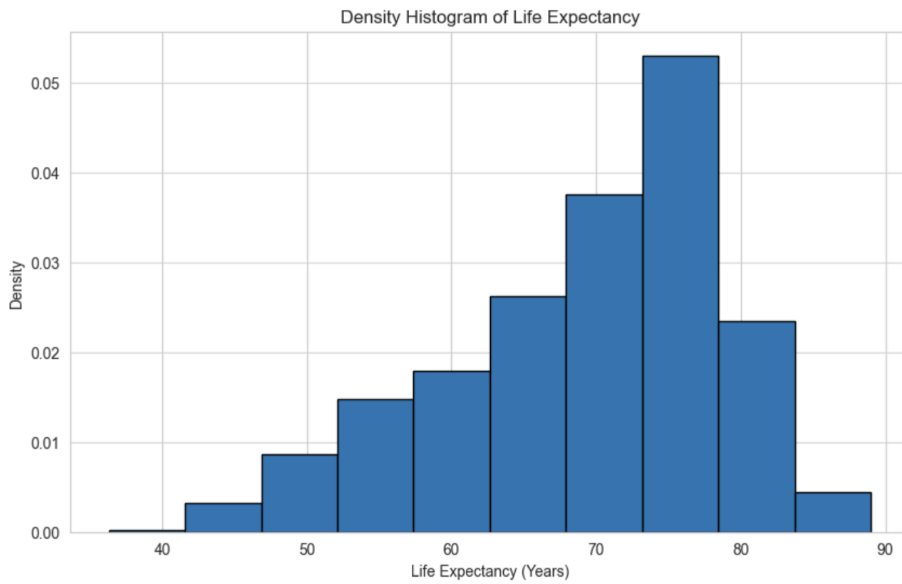
The 7 stages of data science lifecycle:

1. Define the problem
    a. Our big picture problem looks at how socioeconomic factors, such as poverty, environmental health, and fertility, interact with each other to influence life expectancy around the world. At first, we looked at different variables such as the height for men and women, but decided to take those datasets due to not being able to join those datasets with our other datasets. It may also be important to bring this project further by including things such as education, ethnicity, etc.

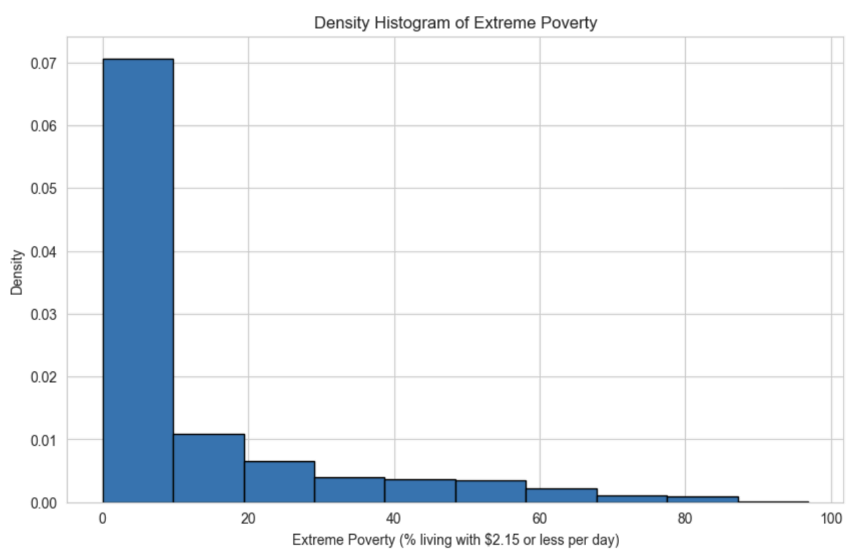
2. Data collection
    a. Our main data set with life expectancy and its associated variables were downloaded from Kaggle, but the data was collected from the World Health Organization. Our 4 other additional datasets on poverty, economic inequality, sanitation, and fertility rate come from Our World in Data.
    b. The merged dataset consisted of up to 15 years of data, spanning from 2000 to 2015, for a total of 101 countries.
    c. The features of interest in our dataset include 'Life Expectancy', ‘Extreme Poverty’, ‘Fertility Rate’, ‘BMI’, ‘Safe Sanitation’, and ‘Population’.
    d. Some countries may not have all the information collected for each feature; therefore, some countries were dropped from the overall dataset. Additionally, the data from our cleaned dataset only spans from 2000 to 2015, which means the relationships observed in the model may not be generalizable for later periods due to evolving global conditions and advancements.

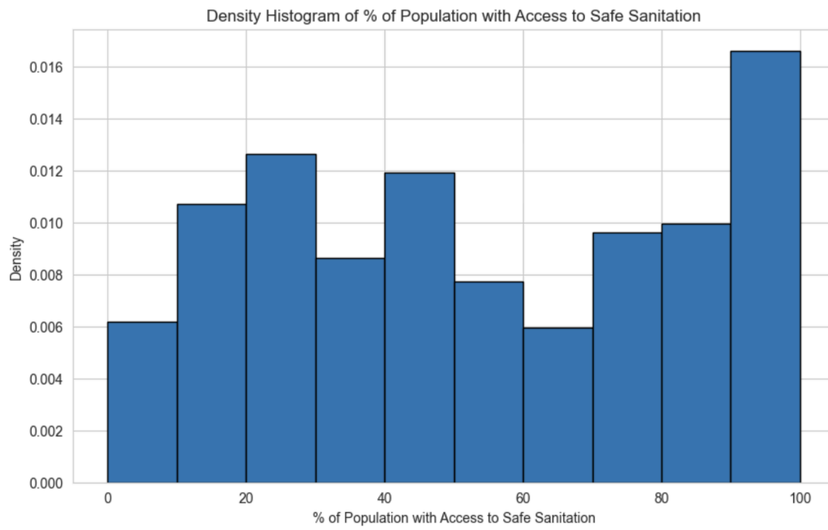
3. Data preparation
    a. Our starting dataset is life expectancy, which includes 2938 rows and 27 columns. We discussed which features we wanted to keep for our predictive modeling and dropped unnecessary columns. Features we wanted to keep are related to environmental health indicators and socioeconomic factors. We then merged the other datasets with the life expectancy dataset on the ‘Country’ and ‘Year’ columns and dropped those columns for modeling purposes. After merging with the four other datasets, the overall data was narrowed to 817 rows with 6 columns, after dropping rows that contained any NA values.
    b. The features we used for our modeling are all numeric data types, therefore, no other data preparation was needed other than merging the datasets and deleting unneeded columns.

4. Data exploration
    a. From the summary statistics, we can see there are significant global disparities in socioeconomic and health factors. Life expectancy averages around 69 years and has a maximum value of 89 years. Its minimum and maximum values do vary widely, which could highlight differences among living conditions. There are also some countries, however, such as Afghanistan, that have lower life expectancies, which indicates that there may be deeper issues.

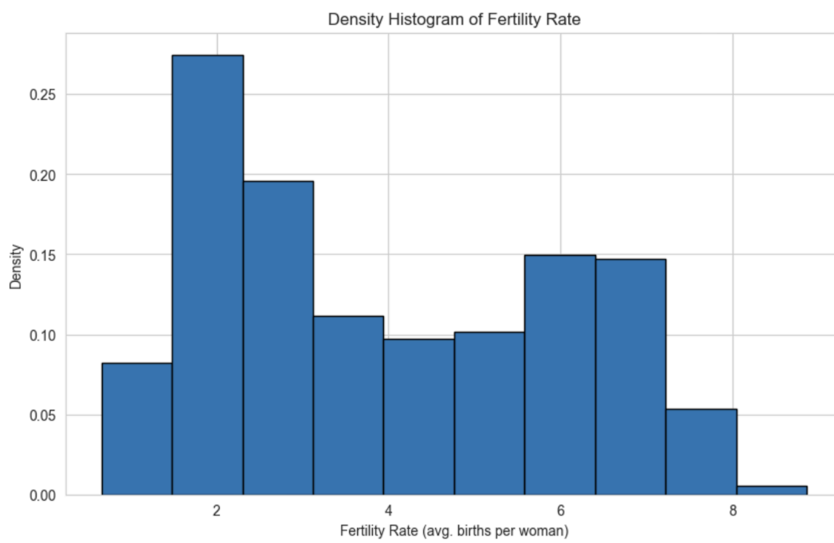


    b. Additionally, access to safe sanitation averages around 53%, but the values range from 0% to 100%, which shows that there are countries with drastically different levels of sanitation.
    c. Poverty seems to vary a lot; the average amount of people living in poverty is 11.6%, but most countries seem to be close to 0%, while others are near 97%. Safe sanitation seems to be similar, with some countries being close to 0% and 100%, but have a different average of 53.4%.





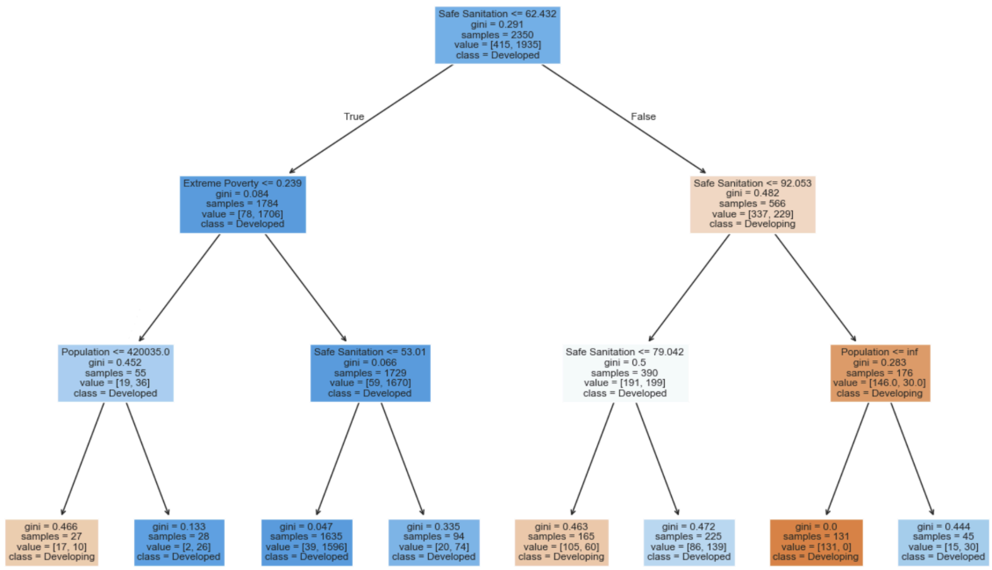
d. Fertility is another factor with an average of about 4 children per woman and a range from 0.66 to 8.86. Higher fertility seems to show up in countries where the life expectancy is lower. We also analyzed height, with women averaging 157 cm and men averaging 168 cm. This might not directly impact someone's lifespan, but it is still something worth considering, as other factors may correlate with it.



e. We thought it would also be interesting to look further into how developed and developing countries compare in these variables, and we found that developed countries tend to have higher life expectancy and better living conditions (less % of people living in extreme poverty and increased access to safe sanitation) than still-developing countries. We also built a Decision Tree Classifier that used sanitation as the top-level feature, poverty, and population as second-level features to determine if a country is developed or developing. This confirms that sanitation is a key indicator for our modeling.

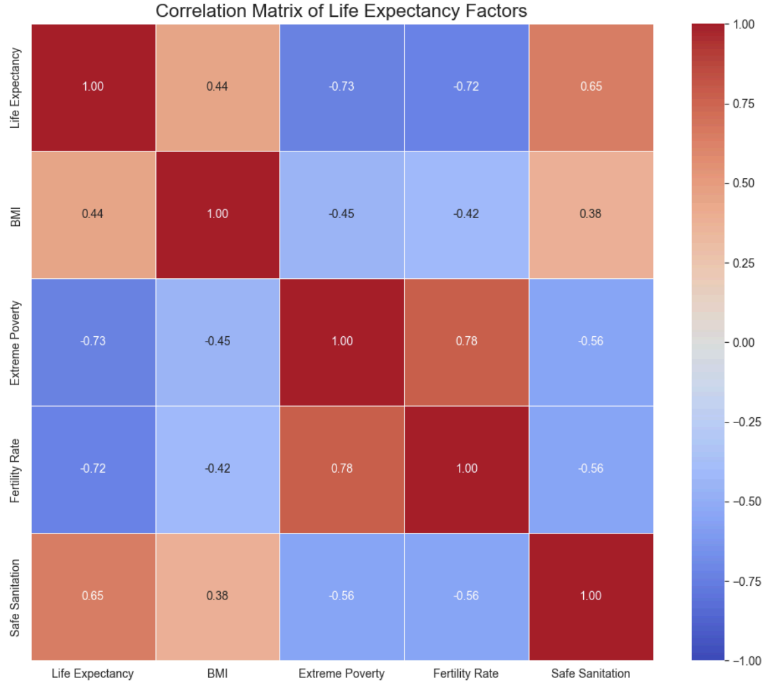
	Life Expectancy	BMI	Extreme Poverty	Fertility Rate	Safe Sanitation
Status					
Developed	79.450456	52.882067	0.605095	1.534951	83.421449
Developing	72.163525	42.873975	10.947936	2.599049	48.283926

Decision Tree for Developed vs Developing

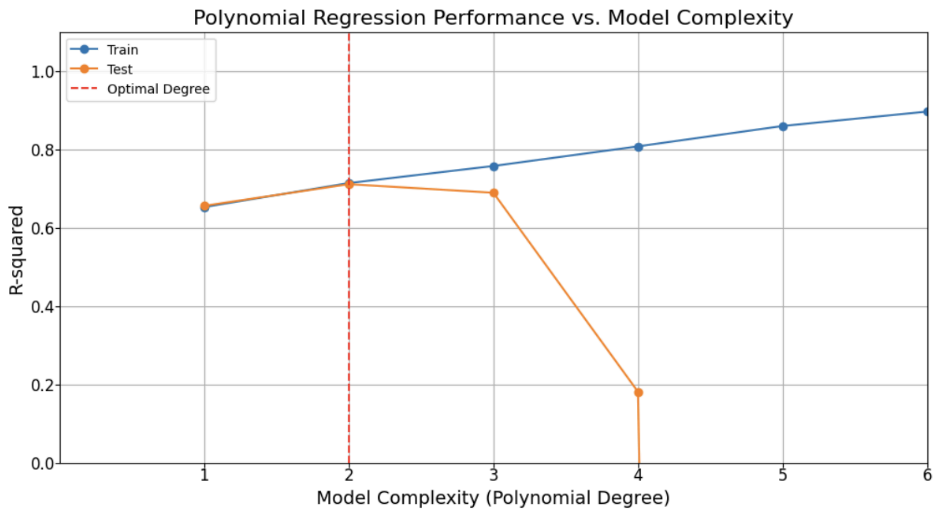


f. To inform our model choice, we further explored the correlations between our chosen variables ('Extreme Poverty', 'Fertility Rate', 'BMI', 'Safe Sanitation') and 'Life Expectancy' using scatter plots and correlation matrices. We observed a negative correlation between 'Extreme Poverty' and 'Life Expectancy', suggesting that higher poverty levels are associated with lower life expectancy.

Correlation Matrix of Life Expectancy Factors



5. Model building
    a. A linear regression model was initially used to see if there were any linear relationships between the predictor variables (*BMI*, *Extreme Poverty*, *Fertility Rate*, *Safe Sanitation*) and the target variable, *Life Expectancy*. The model produced an R^2 of 0.66 and a RMSE of 4.28. Looking at the scatter plots there were some non-linear relationships shown so we explored polynomial regression models with different degrees to see if a polynomial regression model would be a better fit. After our exploration, we found that a polynomial regression model with a degree of 2 was the best at fitting the data.

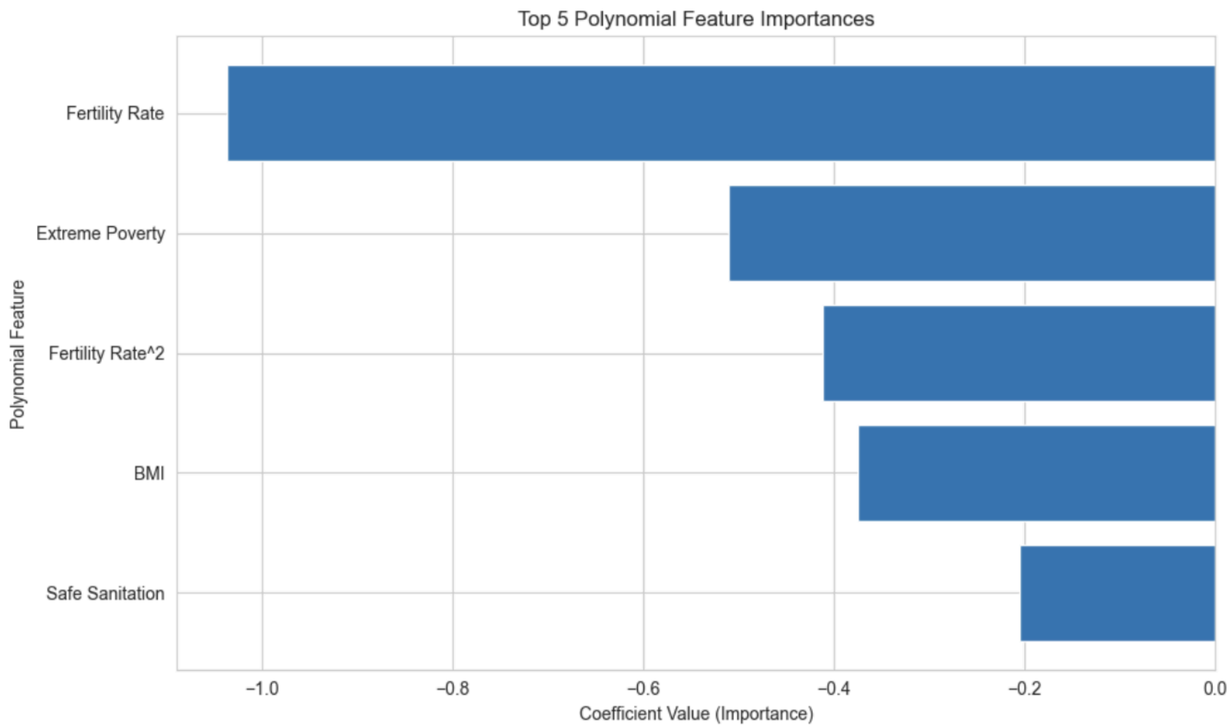


    b. The features 'Extreme Poverty', 'Fertility Rate', 'BMI', and 'Safe Sanitation' were selected based on their relevance as key socioeconomic, environmental, and health factors that are expected to influence lifespan. Below is a table explaining each feature:

Extreme Poverty	Share of the population living in extreme poverty ($2.15 per day)
Fertility Rate	Average births per woman
BMI	Average Body Mass Index of entire population
Safe Sanitation	Share of the population using safely managed sanitation facilities

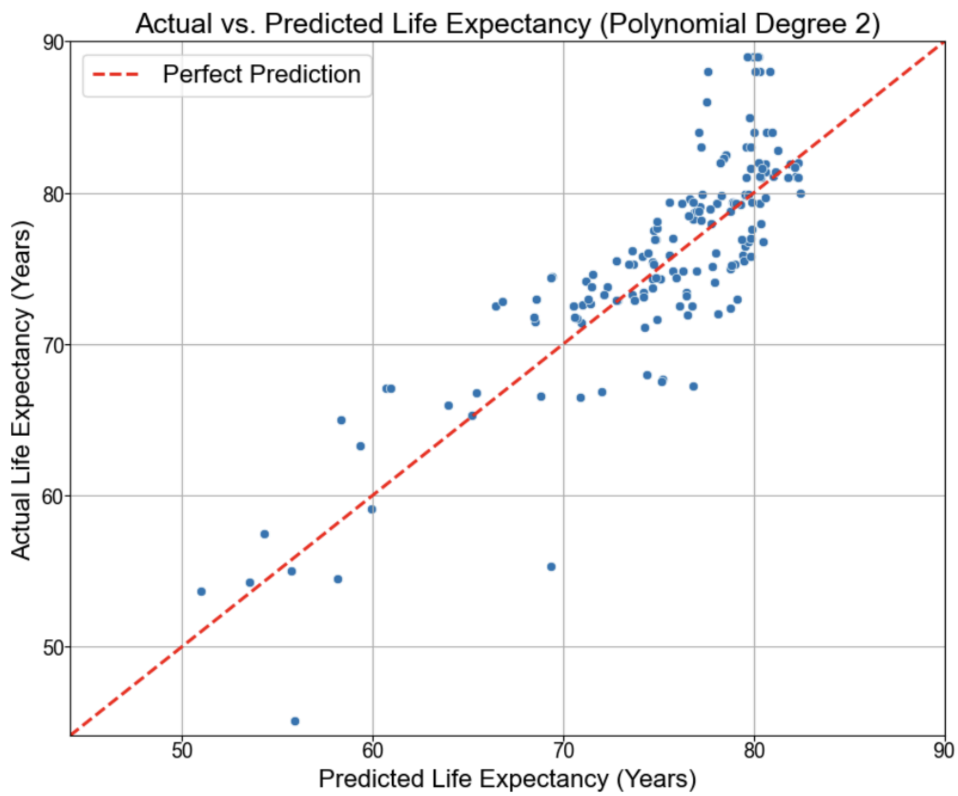
6. Model evaluation
    a. The question we investigated was “How much do factors like extreme poverty, inequality, sanitation access, and fertility rates help explain differences in life expectancy around the world?” In addition, we also wanted to know which factors contributed more to life expectancy, if any.
    b. Because of this question, we created a model to determine the importance and effects of 'Extreme Poverty', 'Fertility Rate', 'BMI', and 'Safe Sanitation' on 'Life Expectancy'.

c. Our final model was a polynomial regression model with a degree of 2 that yielded an R^2 of 0.71 and an RMSE of 3.86. From this model, we were then able to determine the importance of each feature and how it affects life expectancy.



d. Fertility rate was seen to have the largest negative association with life expectancy. As the fertility rate increases, life expectancy tends to decrease, after accounting for the other variables. Extreme poverty, a key socioeconomic factor, also shows a strong negative impact, which highlights that inadequate resources and limited access to healthcare significantly reduce lifespan. The negative coefficient value for BMI also suggests that a higher BMI may be correlated with other health issues that negatively impact life expectancy, though to a lesser extent than the previous variables.

e. The data was split into training (80%) and testing (20%) sets to evaluate the model's generalization performance. A scatter plot of actual vs predicted life expectancy was also used to help visualize the model's accuracy. The plots were generally close to and aligned well with the red line of perfect prediction. However, the plots do start to deviate from the line towards the right tail, which could indicate there were more complex relationships that could not be captured by the model.



7. Model deployment
    i. This model could be used by international organizations (WHO, World Bank, etc.) to identify countries where interventions are most needed to improve life expectancy. For example, countries with high rates of poverty or fertility rates since these factors were found to be associated with lower life expectancies.
    ii. Models could build upon our model into public health systems to see how life expectancy might change based on additional or fixed variables
    Someone would need to prove their and interpret their own results rather than just relying on the outputs. Being able to distinguish between causation and correlation is necessary. Additionally, it is good to be cautious when implementing interventions in the real-world, that are based on the findings of models.
    iii. Since it's hard to find quality data for some regions of the world, the information we did find could misrepresent a region or skip some regions entirely.